Murder at Hiney Hall

Evidence cards

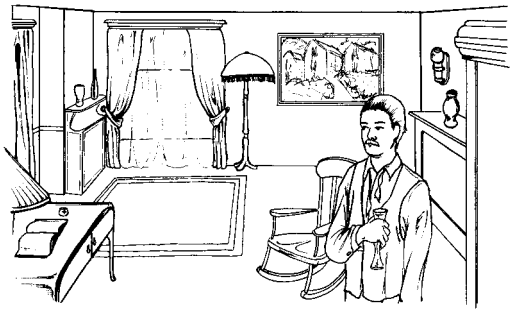
1 *Sergeant Perry, officer in charge at the scene.* I was summoned to the house and arrived at 10.30 p.m. Mr Alfred Dunnet was dead on the floor of the study. He had been shot in the back and his head had been chopped off. There was a great deal of blood on the carpet, but no sign of the head. The lock on the door of the study showed signs of having been forced in. The window was open.



2 *Mr Roger Hare, the butler.* I have been with the family for three years. I had been reading a murder mystery novel in my room since eight o'clock when I heard a shot at about nine fifteen. I immediately came downstairs and found the study door open with Mr James standing over Mr Alfred's body. He had a smile on his face. I suggested that we should telephone for the police, and he did so. The window was open. As Mr James finished on the telephone, Mr Toby walked in, covered in blood. I went and called for Miss Virginia and sent the maid for Mrs Dunnet, who had gone to bed early.



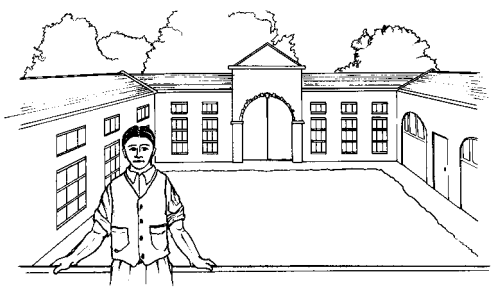
3 *Mr James Dunnet, brother of Mr Alfred Dunnet.* I was in the stables a bit after nine o'clock when I heard a single shot from the house. I rushed back inside and went to the study to find out from Alfred what was going on – the study is where Alfred keeps – I mean kept – his guns. There he was, stone dead on the floor. I was still standing there thinking that someone had done me a whacking great favour a few minutes later when Hare rushes in, takes one look and starts blathering about calling the police, so I did. At least with him gone I won't always be going begging for a few pounds – he inherited everything you know so I have to ask him whenever I need a few bob.



4 *Mrs Alice Dunnet, wife of Alfred Dunnet.* I was in bed asleep at the time it all happened. I must confess that I had gone to bed with a headache after the steaming great row that Alfred and Toby had this evening. When Hare came and woke me up I came down to the study. I saw Alfred in his favourite waistcoat that was only washed the other day, horribly mutilated on the floor, the poor man. You must catch the fiend that did this!



5 *Mr Toby Dunnet, son of Mr and Mrs Alfred Dunnet.* I had had a rattling row with father at tea time, all about which university he wants me to go to. Aberdeen simply won't do, just because he went there. I went out hunting to get some space, brought down a stag but it was a poor shot, didn't kill the beast outright. I had to put the poor beast out of it's misery with a shot to the head, which is why I'm covered in all this blood. Poor old father, eh? What bad luck.

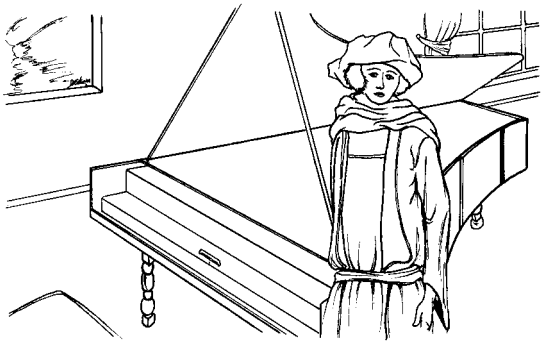


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Murder at Hiney Hall continued

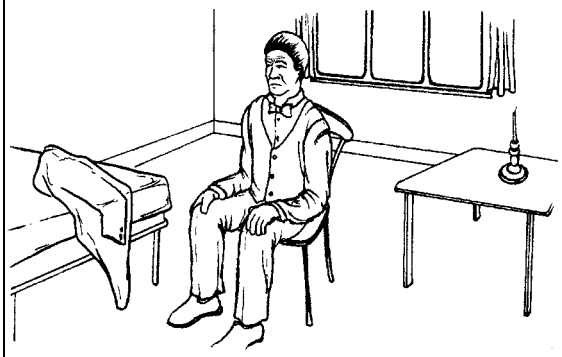
6 *Miss Virginia Dunnet, daughter of Mr and Mrs Alfred Dunnet.* I was alone in my room, playing Cluedo, so I didn't hear a thing. The first I knew was when Roger Hare came and told me the dreadful news.



7 *Miss Ethel Bragg, maid to the Dunnets.* About half past nine Mr Hare comes knocking on my door saying Mr Alfred has been murdered and I must wake Mrs Dunnet. I was a bit taken aback – he doesn't normally come knocking on my door, and I had said goodnight to him at nine o'clock when I came up to bed. Poor Mr Alfred. So terrible.



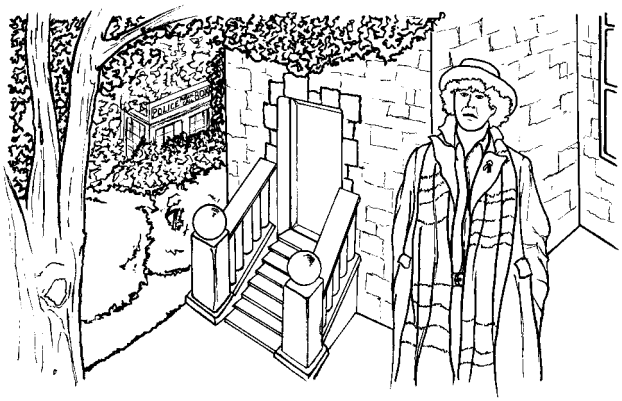
8 *Mr Roger Hare.* Second interview. I lied. I wasn't in my room. I was with Miss Virginia in her room, playing some silly game. We love each other, and would get married except that Mr Alfred had forbidden it!



9 *Miss Virginia Dunnet.* Second interview. Oh, did I forget to mention that Roger was with me? Silly me. Yes he was there all the time until quarter past nine when he said he had heard a noise and went off to investigate. Isn't he brave?



10 *Doctor Algernon Hoo.* You know I can't give you much information until I've examined the body properly, but I can tell you that this is the body of a middle-aged man, and he has been dead for at least two days.



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Identifying solids, liquids and gases

Aim

To identify solids, liquids and gases.

Equipment

- different materials

What to do

1. Look at each sample in turn.
2. Decide whether it is a solid, liquid or gas.
3. Write the name of the sample and your decision in a results table similar to the one below,

Results

Substance	Solid/Liquid/Gas?	Correct answer

Questions

1 Which were there most of, solids, liquids or gases?

2 How did you know which ones were liquids?

3 How did you know which ones were gases?

4 How did you know which ones were solids?

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Experiments with solids, liquids and gases

G2
Core



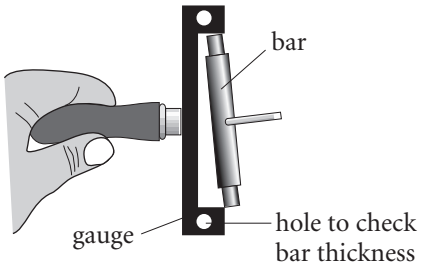
Aim

To explain why solids, liquids and gases behave differently.

What to do

1. Go to each work station.
2. Carry out the instructions with each experiment.
3. Answer all the questions before moving on to the next work station.

Work station 1: Bar and gauge



Questions

1 Did the length of the bar fit along the gauge when it was **a** cold **b** hot?

2 Did the thickness of the bar fit along the gauge when it was **a** cold **b** hot?

3 What has happened to the length and thickness of the bar?

Work station 2: Dissolving crystals

Questions

4 Use a hand lens to help you draw the crystals before you put them into the water.

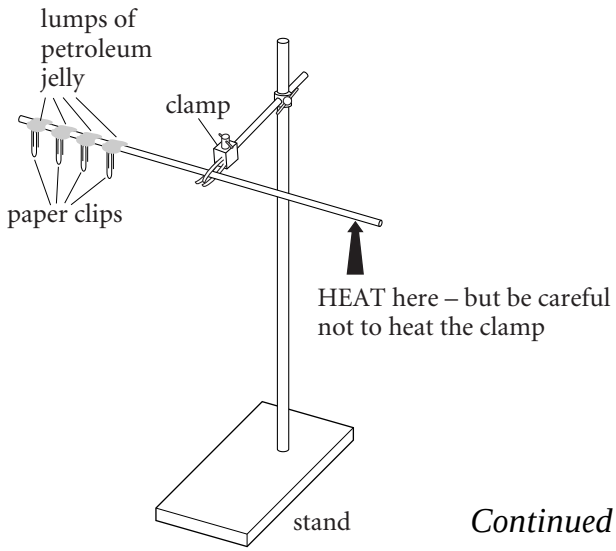
5 Draw what happened in each beaker by making sketches every 30 seconds or every minute.

6 What happened to each crystal that you put into the water?

7 What difference did the hot water make?

Work station 3: Rod and paperclips

Write down any changes that occur and when they occur.



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Experiments with solids, liquids and gases

G2
Core



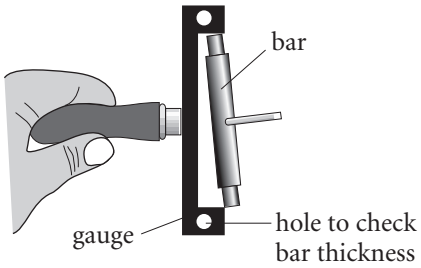
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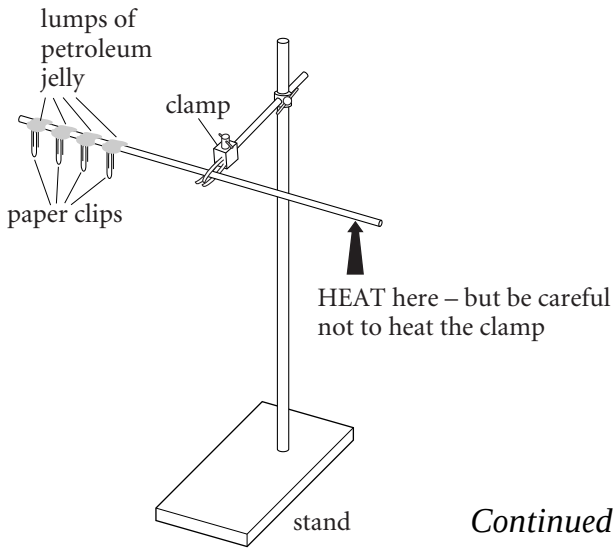
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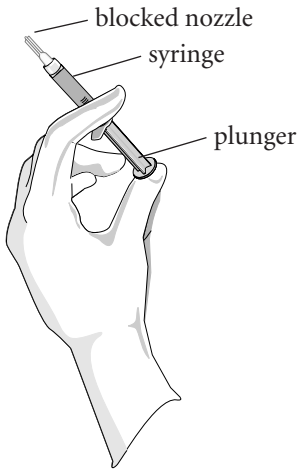
Experiments with solids, liquids and gases continued

Work station 4: A perfume bottle

Question

8 Describe what happened to the smell of the perfume as you got closer to the bottle.

Work station 5: Syringes



Questions

9 Could you press the plunger in when there was a solid in the syringe?

10 Could you press the plunger in when there was a liquid in the syringe?

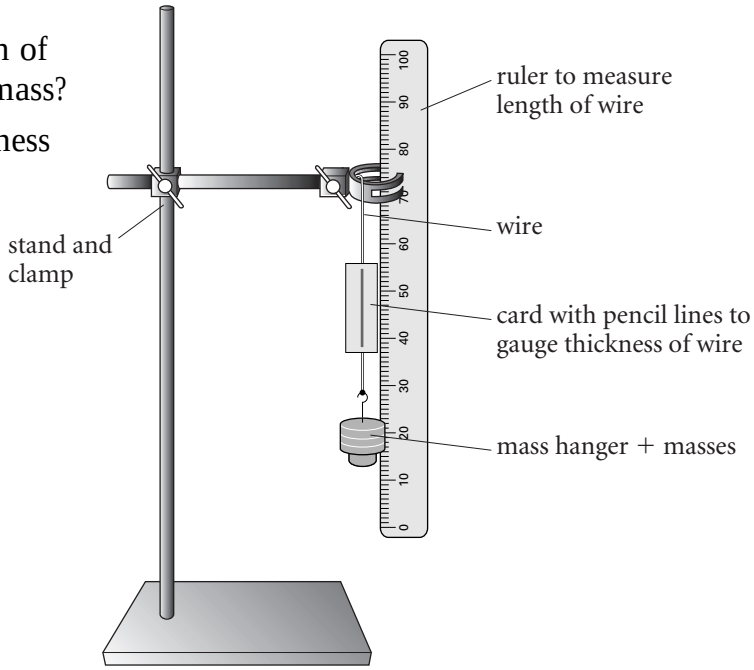
11 Could you press the plunger in when there was a gas in the syringe?

Work station 6: Stretching a wire

Questions

12 What happened to the length of the wire as you added each mass?

13 What happened to the thickness of the wire during the test?



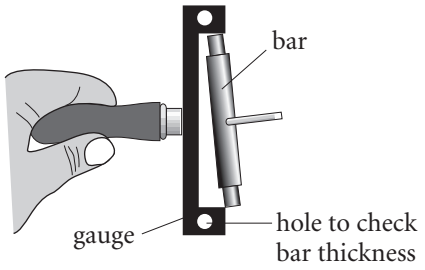
Finally

When you have completed the activities, try to write down an **explanation** of what has happened in each experiment. Discuss your ideas in your group and then prepare a short presentation to explain your ideas to your class.

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Workstation cards

Work station 1: Bar and gauge



1 Take a bar and gauge from the cool area.

2 Clamp the gauge.

3 Check that the bar fits in the gauge, both along its length and across its thickness.

4 Heat the bar only.

5 Carefully try to fit the bar into the gauge.

6 When you have finished, put your bar and gauge in the 'hot' area.

Work station 2: Dissolving crystals

1 Half fill one beaker with cold water from the tap and the other beaker with hot water from the kettle.

2 Leave the water still for at least 30 seconds while you read questions 1 and 2.

3 Use tweezers to take two large crystals from the dish and put one crystal into each beaker of water.

4 Watch the crystals carefully for 3 minutes.

Work station 3: Rod and paperclips

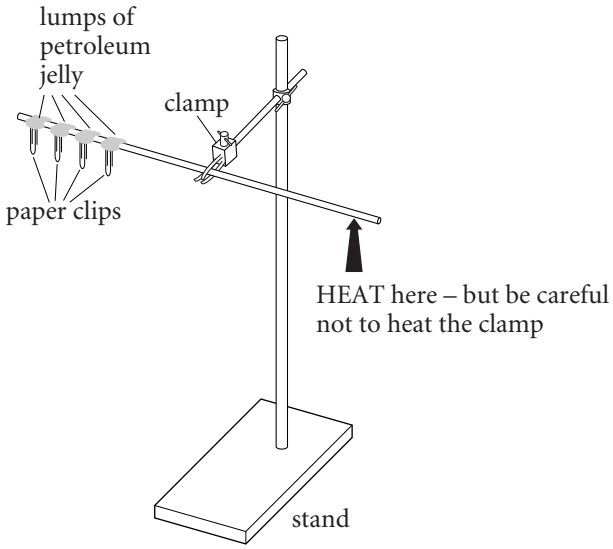
1 Clamp the rod exactly as in the diagram.

2 Use a spatula to put four blobs of petroleum jelly on the rod as shown.

3 Gently stick a paperclip into each blob of jelly so that the paperclip hangs underneath the rod.

4 Put the Bunsen burner underneath the end of the rod away from the paperclips and heat the rod.
Start the stopwatch as you do this.

5 Watch the rod, jelly and paperclips carefully, and write down any changes that occur and when they occur.



Continued ⟶

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Workstation cards continued

Work station 4: A perfume bottle

1 STOP one metre from the bottle.

2 Check that the top has been taken off the bottle.

3 Sniff the air. Can you smell the perfume?

4 Move your nose 10 cm towards the bottle at a time and sniff. Write down the distance from the bottle at which you could first smell the perfume.

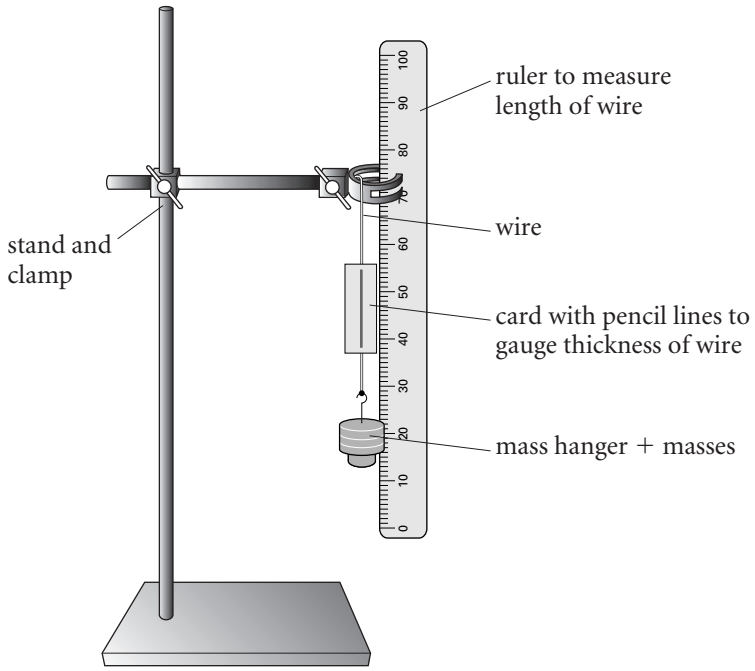
5 Keep moving towards the perfume bottle 10 cm at a time and sniffing.

Work station 5: Syringes



Using the grip shown in the diagrams above ONLY, try to push the plunger of each of the three syringes. Be careful not to press so hard that you hurt yourself or damage the syringe!

Work station 6: Stretching a wire



1 Lay the wire on the piece of card, with one edge of the wire against a ruler. Draw a pencil line down the side of the wire. Move the ruler to the other side of the wire and draw a line down that too. Check that the two lines you have drawn are side by side and have a gap between them the same as the thickness of the wire. If not, try again!

2 Wrap one end of the wire around the clamp so that it cannot slip.

3 Wrap the other end of the wire around the hook of the mass hanger.

4 Use the ruler to check the length of the wire and the guide lines to check the thickness.

5 Add one mass to the hanger.

6 After you have carefully added the mass, check the wire to see if it has changed.

7 Keep adding masses one at a time and measuring, until the wire snaps.

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Classifying solids, liquids and gases

Aim

To classify materials on the basis of their properties.

Equipment

- pens and large sheets of paper
- access to a variety of solids, liquids and gases

What to do

Part one: making the key

1. Think of a question that *might* help you to tell solids, liquids and gases apart.
2. Write the question down.
3. Look at some materials, and answer your question.
4. If your question works – that is, the answer to the question is useful in deciding whether a material is a solid, liquid or gas – write it into a list of useful questions.
5. Repeat steps 1 to 4 until you have a good list of useful questions, that can sort out solids and liquids, solids and gases, liquids and gases.
6. Use your list of questions to make a draft key that can classify any material as a solid, liquid or gas.
7. Try your draft key out to make sure it works. You must make sure you use the key questions, rather than just looking at the material and saying, for example 'of course it is a solid'. If the key doesn't work, redraft it and test it again.
8. When you have got a key that works, make a neat copy of it.

Part two: using the key

9. Use your key to classify one of the 'special' materials that your teacher has.
10. For each substance, record its name and whether it is a solid, liquid or gas.
11. If you find any materials that your key can't classify, write them in a separate column.

Questions

1. Does your key work for all solids?
2. Does your key work for all liquids?
3. Does your key work for all gases?
4. Did you find any substance that your key did not work with?
5. Compare your classification with another group. Do you agree on everything?

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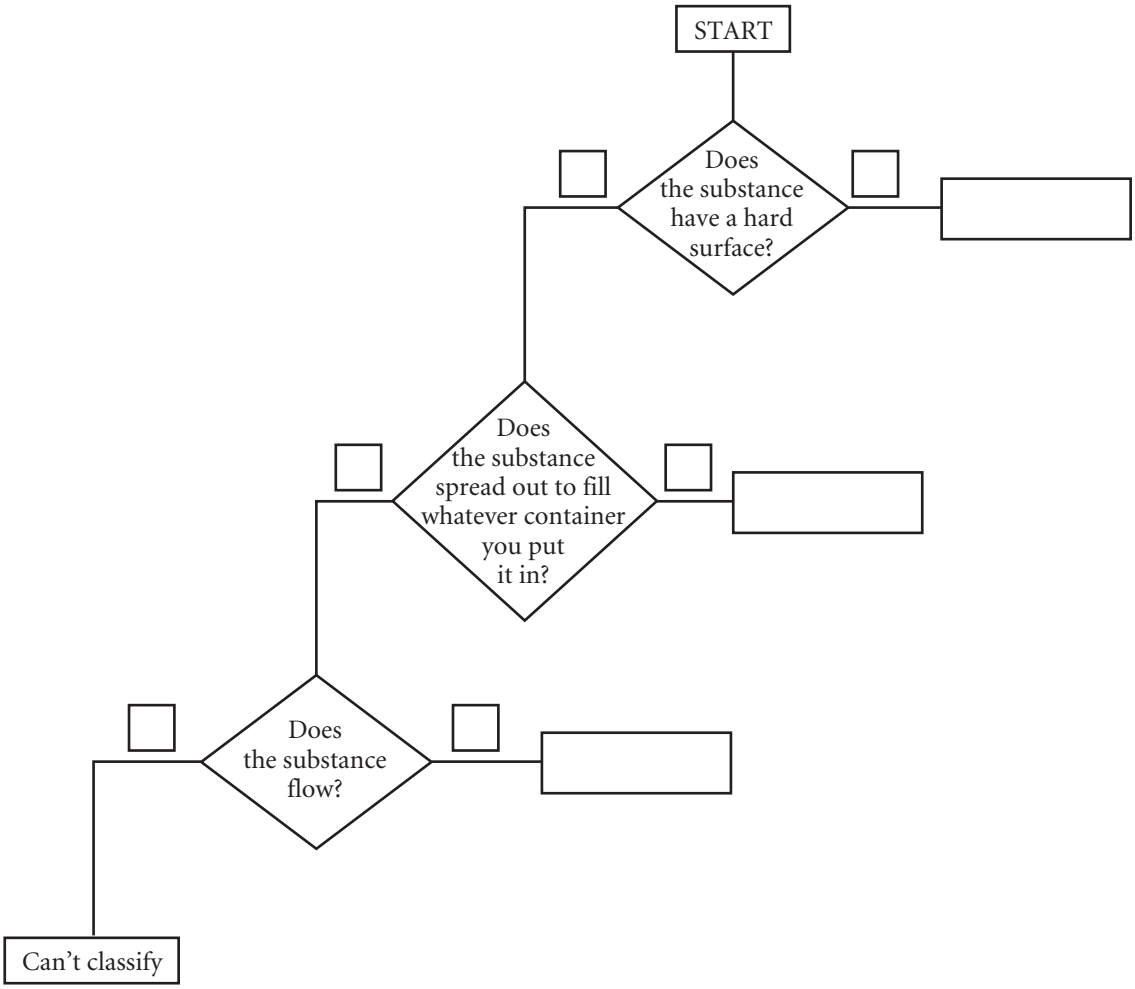
Classifying solids, liquids and gases

Aim

To draw up a key to identify solids, liquids and gases.

What to do

1. Below is a blank key. Complete the key by filling in ''solid'' ''liquid'' and ''gas'' at the right points and ''Y'' or ''N'' in the box on each of the branches from the questions.



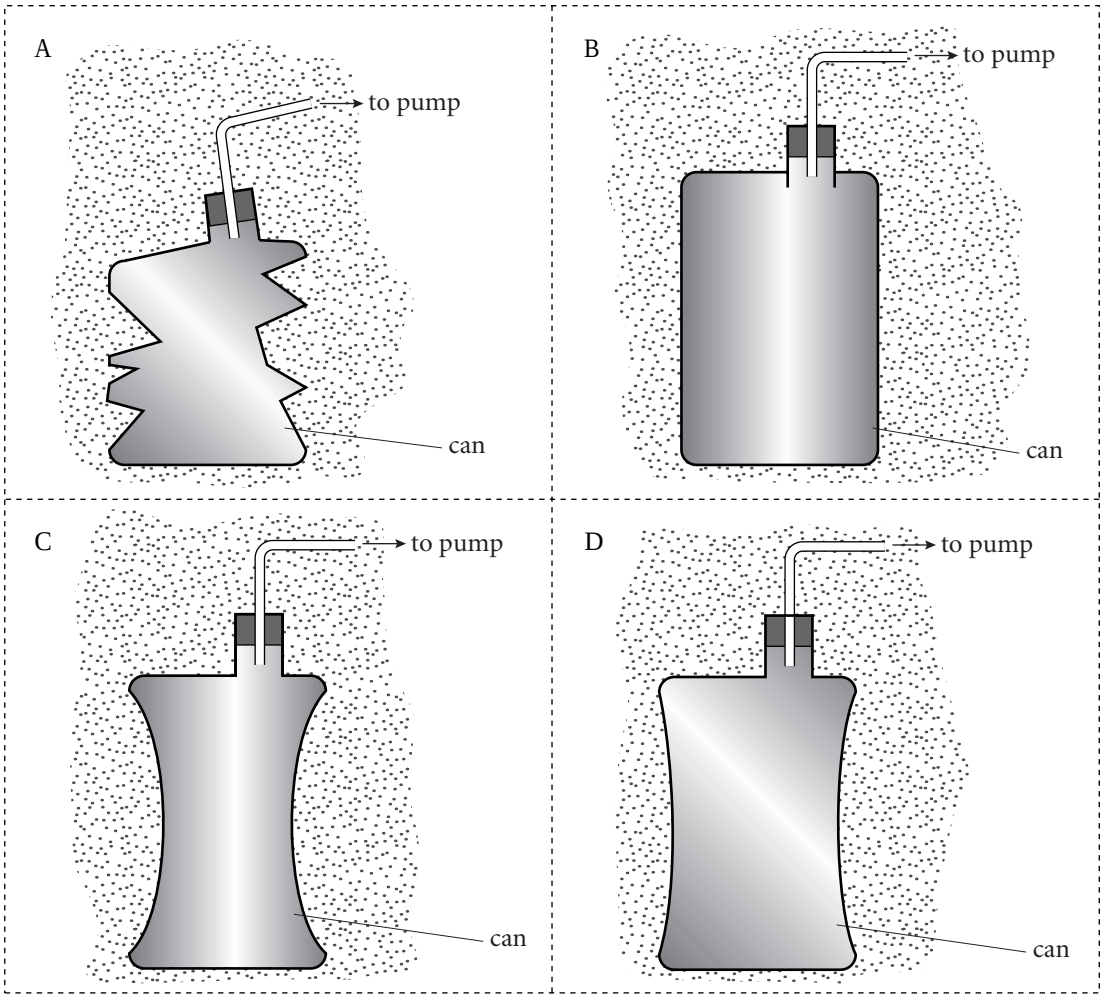
2. Now use your key to classify the materials around the laboratory.

Name of substance	Solid	Liquid	Gas	Can't classify

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The collapsing can

Organise the diagrams and the labels so that they show the correct sequence of events during the collapsing can experiment.



Labels

1 Can with normal amount of air particles in it.

2 Can with almost no air particles left in it.

3 As the air particles are removed, the air pressure falls.

4 When the air pressure is low inside the can, the walls bow inwards.

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Looking at particles moving

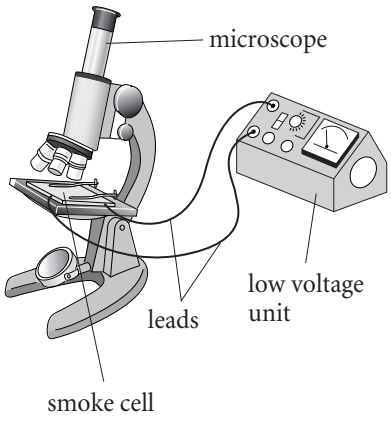
Aim

To observe particles moving.

In 1827 a botanist named Robert Brown was looking through a microscope at pollen floating in a dish of water. To his surprise he saw that the pollen grains were moving around on the surface of the water!

Robert Brown could not explain what he saw, but today we can use our ideas about particles to understand it – the pollen grains were being pushed around when water particles collided with them. The water particles were too small to see with a microscope, but enough of them hitting the pollen grain at the same time could move it. The pollen grains moved *randomly* – they went in all directions – because the water particles hit them from different directions.

In honour of the person who first observed this, the movement of particles became known as 'Brownian' motion.



Equipment

- microscope
- smoke cell
- power source for smoke cell
- smoke generator

What to do

1. Set up the smoke cell and microscope, as shown by your teacher.
2. Carefully focus the microscope until you can see jiggling points of light. These are the smoke particles.
3. Draw two diagrams **a** showing what you saw and **b** showing the possible path of one smoke particle.
4. Explain why the smoke particles move around randomly.

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Is this liquid a mixture?

H1
Core

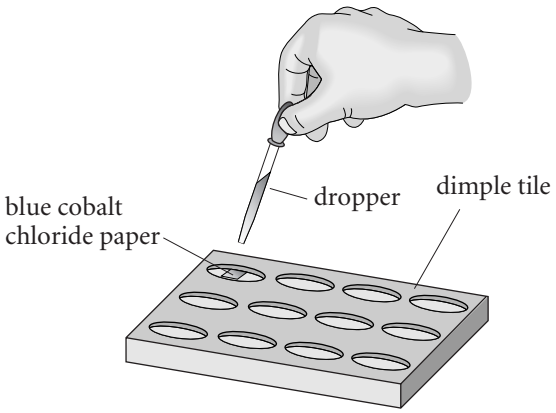


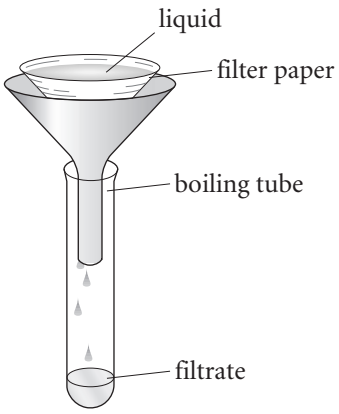
Aim

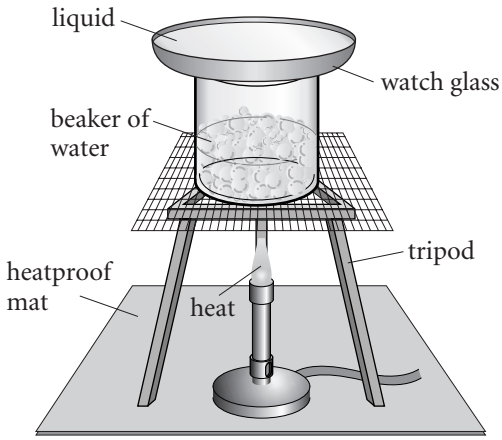
To find out if liquids contain water and to find out if they are pure substances.

Equipment

- various liquids
- cobalt chloride paper
- dimple tile
- droppers
- watch glass
- beaker
- tripod, heatproof mat and gauze
- Bunsen burner
- boiling tube
- filter paper
- funnel







What to do

1. Test each sample in turn to see if it contains water, by putting one drop of the liquid onto a small square of blue cobalt chloride paper. If water is present the paper will turn pink.

2. Test each sample in turn to find out if the liquid is a mixture or not. The first thing to do is to filter the liquid. If there are any undissolved substances mixed in they will stay on the filter paper.

3. Next, you can see if it contains any dissolved substances by evaporating the liquid on a watch glass over a water bath. If the liquid is pure, there will be nothing left on the watch glass when the liquid has all evaporated. If the liquid was not pure, traces of solid will be left on the surface of the glass. These are the other substances that were mixed with the liquid.

4. Fill in the results table in the correct places.

5. Your teacher will tell you what the liquids were. Write this in the third column of the results table. Finally, write in the name of the solid dissolved, where appropriate.

Continued ⟶

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